

$$26.1] \quad f = 7 \times 10^{14}, \quad v = 6 \times 10^5 \text{ m/s}$$

$$K.E = \frac{1}{2}mv^2 = hf - hf_0$$

$$f_0 = -\frac{mv^2}{2h} + f = \frac{9.11 \times 10^{-31} \times (6 \times 10^5)^2}{2(6.63 \times 10^{-34})} = \frac{mv^2}{2h}$$

$$f_0 = f - \frac{mv^2}{2h}$$

$$= 7 \times 10^{14} - \frac{9.11 \times 10^{-31} (6 \times 10^5)^2}{2(6.63 \times 10^{-34})} = 4.5 \times 10^{14} \text{ Hz}$$

$$26.2] \quad \phi = 2.3 \text{ eV} = 2.3 \times 1.6 \times 10^{-19} \text{ J}$$

$$W = \phi = hf_0 = h \frac{c}{\lambda}$$

$$\Rightarrow \lambda = \frac{hc}{\phi} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2.3 \times 1.6 \times 10^{-19}} = 5.4 \times 10^{-7} \text{ m}$$

26.5] The incident light is above the threshold frequency for electrons to be emitted by the plates, so the plates and the leaves become positively charged.

$$26.6] \quad n = \frac{P}{hf} = \frac{10^4}{6.63 \times 10^{-34} \times 9.5 \times 10^5} \\ = 1.6 \times 10^{21}$$

$$26.8] \quad n = \frac{P}{hf}, \quad \lambda f = c \Rightarrow f = \frac{c}{\lambda}$$

$$\Rightarrow n = \frac{P}{h} \frac{\lambda}{c} = \frac{10^3 \times 550 \times 10^{-9}}{6.63 \times 10^{-34} \times 3 \times 10^8}$$

$$= 2.765 \times 10^{21}$$

$$26.10] \quad \lambda = 500 \text{ nm} = 500 \times 10^{-9} = 5 \times 10^{-7} \text{ m}$$

$$(a) \quad E = hf = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{5 \times 10^{-7}}$$

$$\approx 4 \times 10^{-19} \text{ J}$$

$$(b) \quad p = \frac{E}{c} = \frac{4 \times 10^{-19}}{3 \times 10^8} = 1.3 \times 10^{-27} \text{ kg m/s}$$

$$26.14) \quad K_{\text{electron}} = hf' - hf = h(f' - f)$$

$$= 6.63 \times 10^{-34} (2.4 - 1.6) \times 10^{20} = 5.3 \times 10^{-14} \text{ J}$$

$$26.17) (a) \quad E = hf_{\text{max}} = 9 \text{ V} \Rightarrow f_{\text{max}} = \frac{9 \text{ V}}{h}$$

$$f_{\text{max}} = \frac{1.6 \times 10^{-16} \times 50000}{6.63 \times 10^{-34}} = 1.2 \times 10^{22} \text{ Hz}$$

$$(b) \quad \lambda = \frac{c}{f} = \frac{3 \times 10^8}{1.2 \times 10^{22}} = 2.5 \times 10^{-14} \text{ m}$$